

Cross-slot antenna report in D-band

For the antenna topology, Fig. 9 shows a single element cross-slot microstrip patch fed by a microstrip line that is designed and simulated. The patch dimension, slot dimension and the substrate thickness and permittivity is designed for optimum performance. The substrate, having a **relative permittivity of 2.6** has dimensions of **Sub_L**= 4.12mm, **Sub_W**= 2.51mm, and **50 μ m** in thickness.

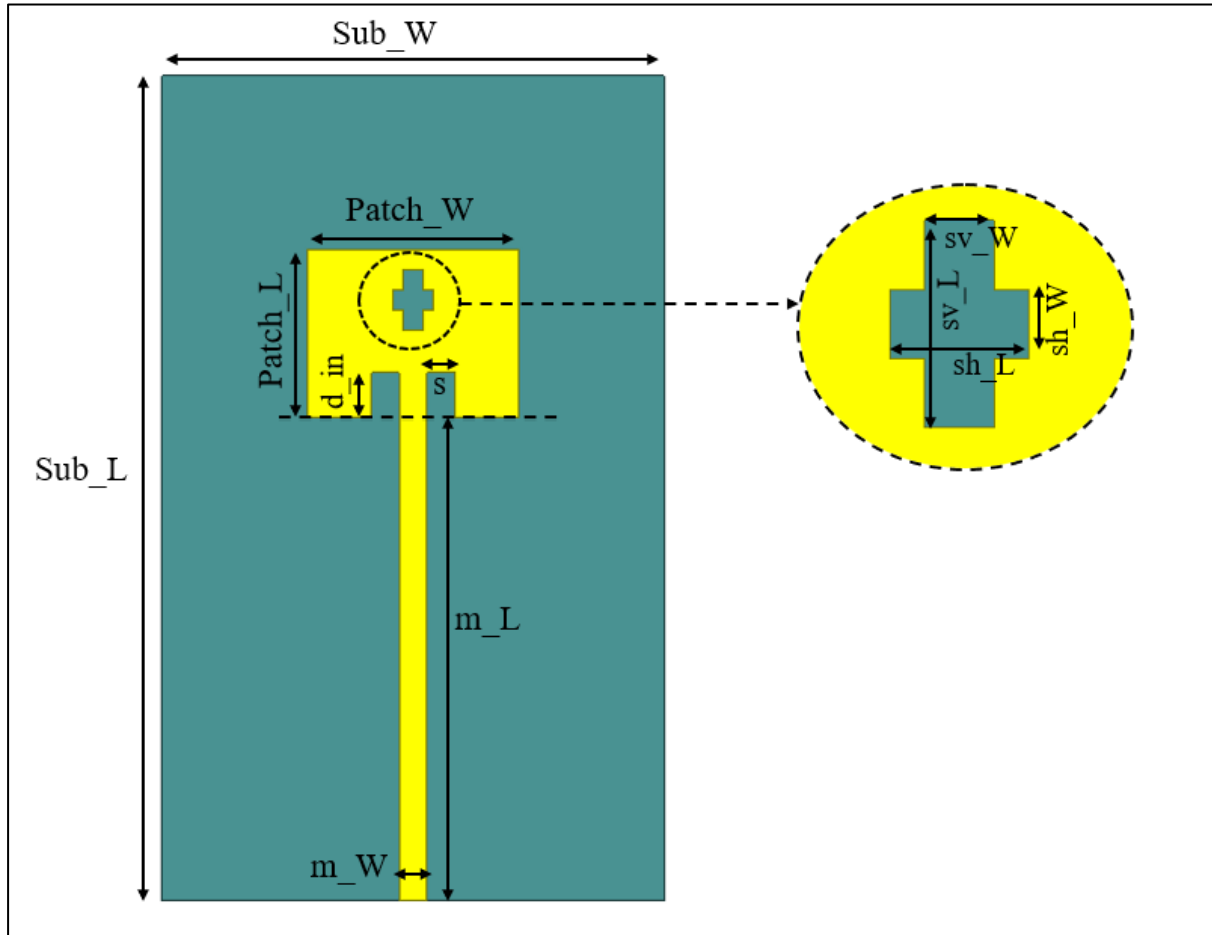


Fig.9. Single element cross-slot microstrip patch antenna with $Sub_L = 4.12\text{mm}$, $Sub_W = 2.51\text{mm}$, $Patch_W = \text{mm}$, $Patch_L = \text{mm}$, $d_in = 0.2\text{mm}$, $s = 0.14\text{ mm}$, $m_L = 2.416\text{mm}$, $m_W = 0.14\text{mm}$, $sv_L = 0.3\text{mm}$, $sv_W = 0.1\text{mm}$, $sh_L = 0.2\text{mm}$, $sh_W = 0.1\text{mm}$.

For the second antenna topology, Fig. 10. shows a 2x2 cross-slot microstrip patch fed by a co-axial probe with **inner and outer radii 0.2 mm and 0.5 mm respectively**. The entire configuration is designed and simulated on a substrate with **relative permittivity of 2.6** and thickness of **50 μ m**.

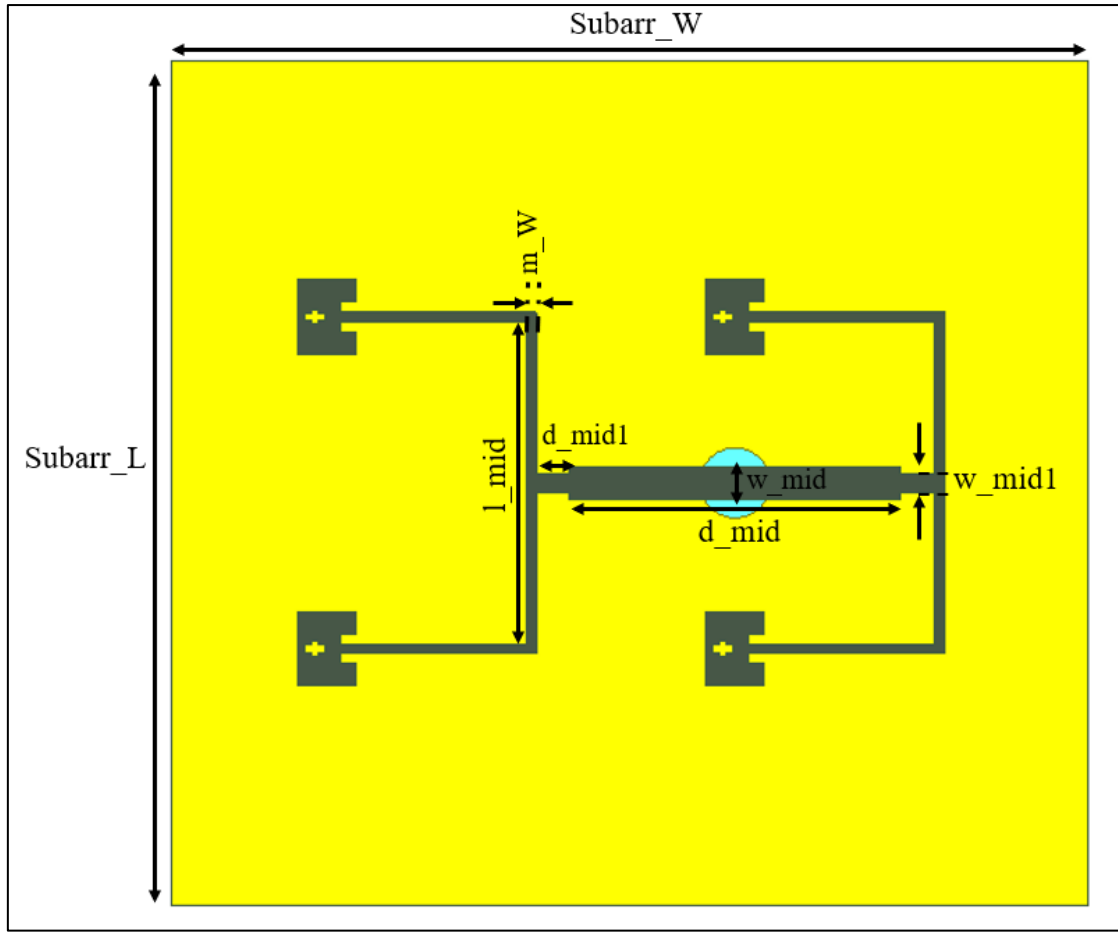


Fig. 10. *2x2 array cross-slot microstrip patch antenna with $Subarr_L = 12\text{mm}$, $Subarr_W = 13\text{mm}$, $d_mid = 4.71\text{mm}$, $d_mid1 = 0.472\text{mm}$, $l_mid = 4.85\text{mm}$, $w_mid = 0.45\text{mm}$, $w_mid1 = 0.259\text{mm}$.*

The figure depicts a cross-slot microstrip patch array antenna that employs the same cross-slot patch design as described previously. The array consists of four identical patches arranged symmetrically on a $12\text{mm} \times 13\text{mm}$ substrate. Each patch has dimensions consistent with the earlier description.

Results for Single element antenna:

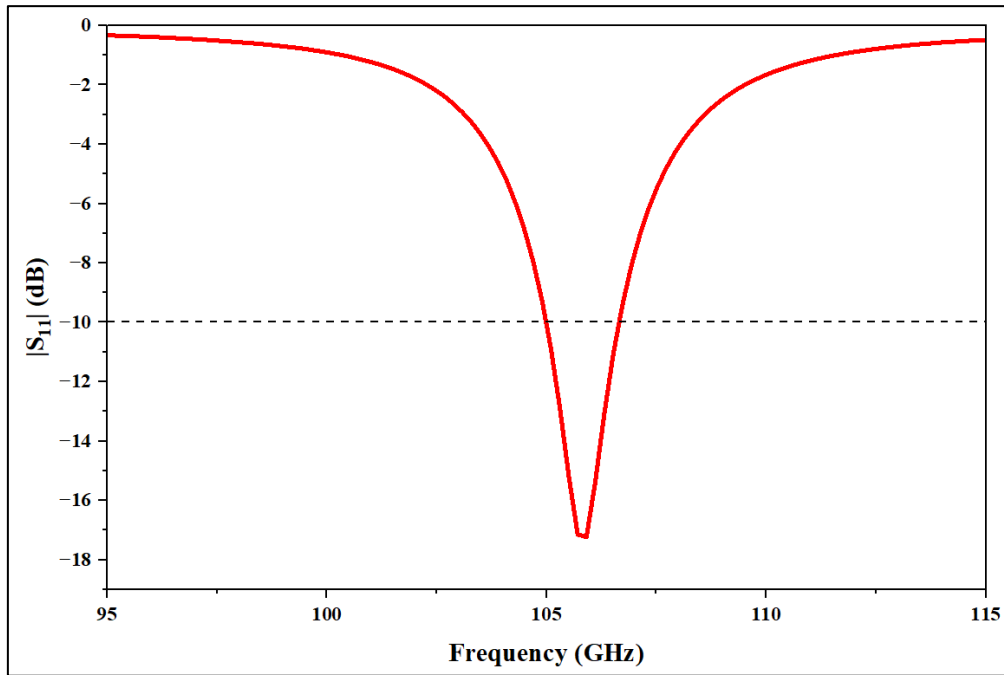
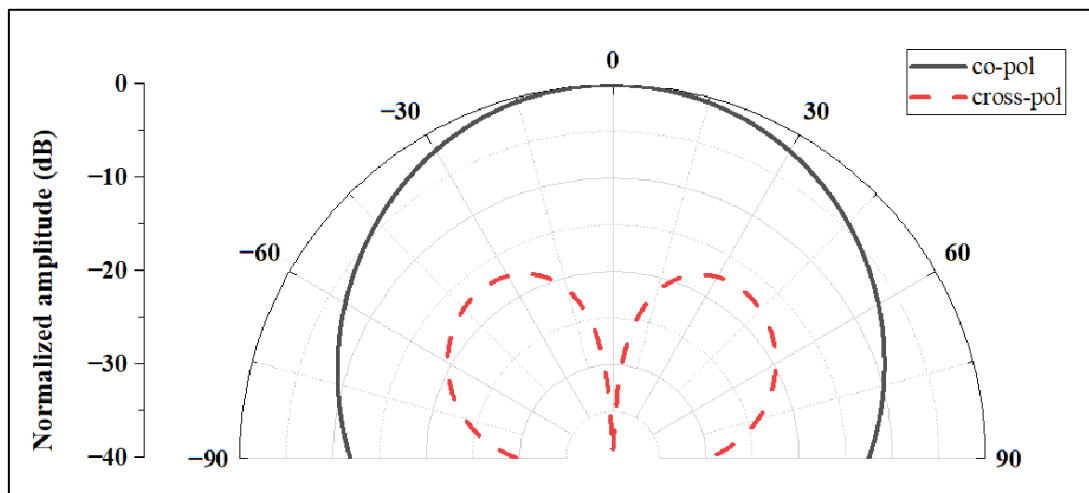


Fig. 11. Reflection coefficient vs frequency of a single element cross-slot antenna.

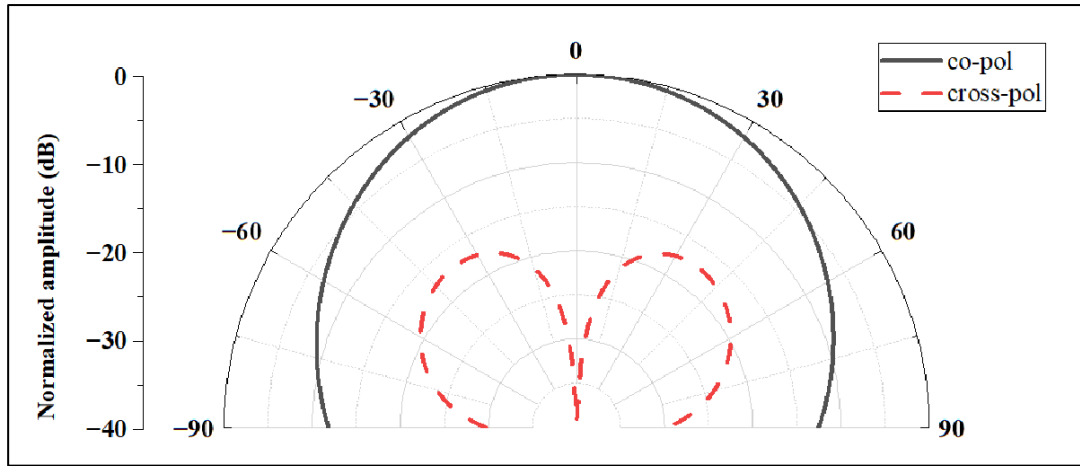
The reflection coefficient curve for the single element u-slot patch antenna, as shown in fig. 11. gives us a maximum dip of **-17.22 dB at a resonant frequency of 105.9 GHz.**

The **percentage bandwidth** is calculated to be **1.7%.**

The radiation patterns at the $\phi = 0^\circ$ and $\phi = 90^\circ$ planes at 105.9 GHz is shown in in Fig. 12.



(a) $\phi = 0^\circ$



(b) $\phi = 90^\circ$

Fig. 12. Radiation pattern of a single element cross-slot antenna at 105.9 GHz (a) $\phi = 0^\circ$ (b) $\phi = 90^\circ$

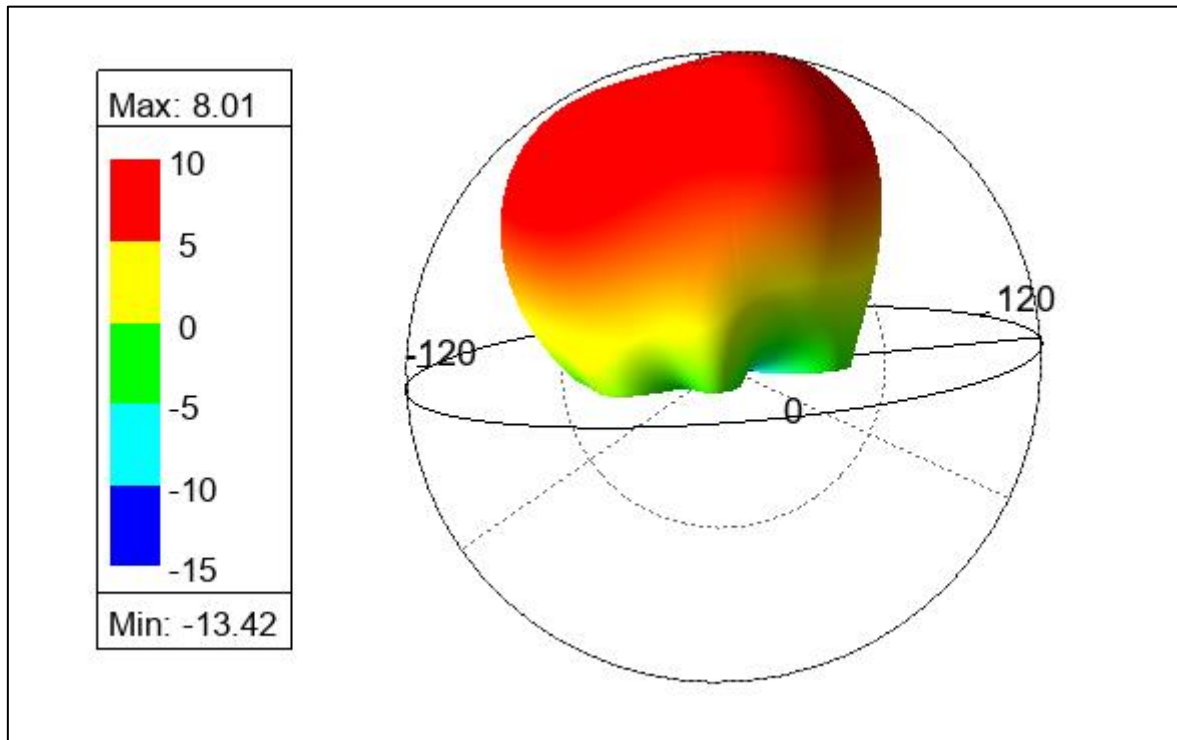


Fig. 13. 3D radiation pattern of single element cross-slot antenna at 105.9 GHz.

The 3-dimensional radiation pattern in figure 9, shows a **maximum gain of 8.01 dB** of the single element cross-slot antenna at the resonant frequency of 105.9 GHz.

Results for array antenna:

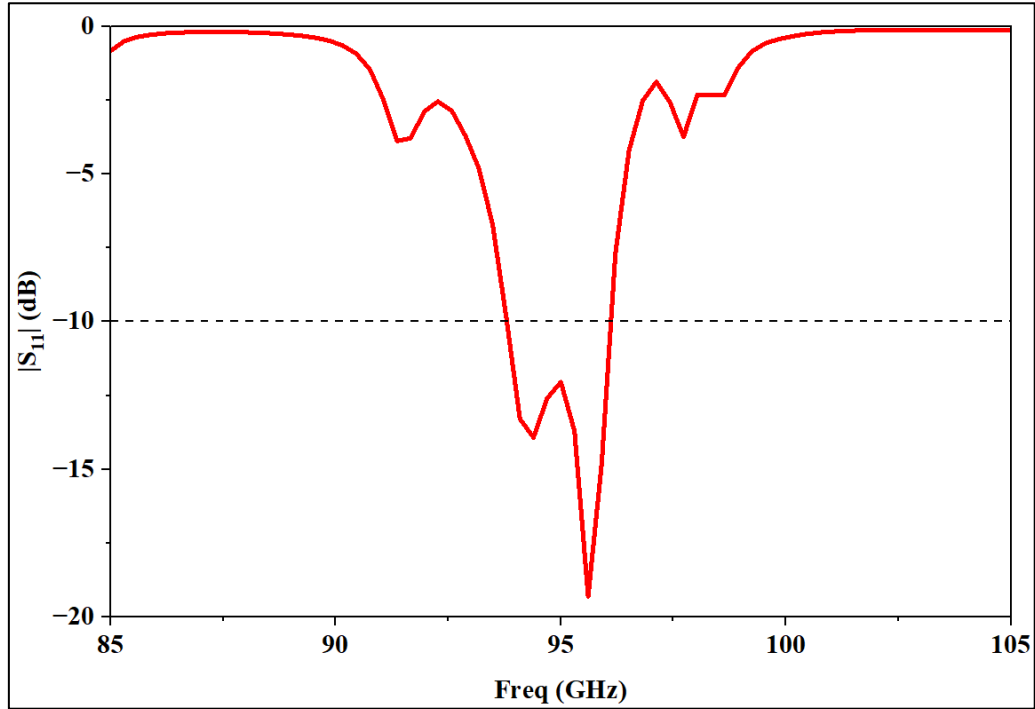
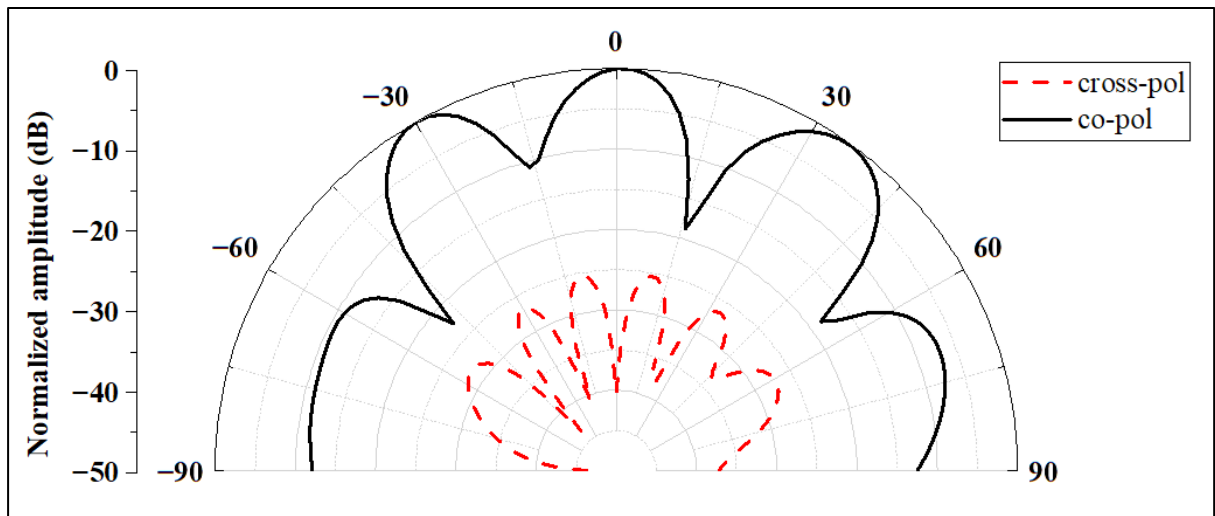


Fig. 14. Reflection coefficient vs frequency of 2x2 cross-slot array antenna.

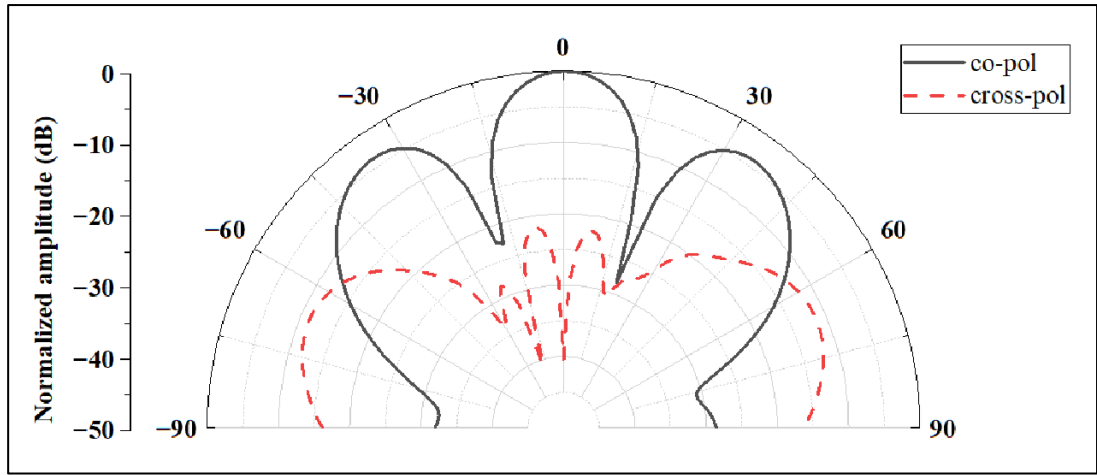
From fig.14, the reflection coefficient curve for the single element u-slot patch antenna, as shown in fig. 11. gives us a maximum dip of **-19.6 dB at a resonant frequency of 96.1 GHz.**

The **percentage bandwidth** is calculated to be **3.12%.**

The radiation patterns at the $\phi = 0^\circ$ and $\phi = 90^\circ$ planes at 96.1 GHz is shown in in Fig. 15.



(a) $\phi = 0^\circ$



(b) $\phi = 90^\circ$

Fig. 15. Radiation pattern of 2x2 cross-slot array antenna at 96.1 GHz (a) $\phi = 0^\circ$ (b) $\phi = 90^\circ$

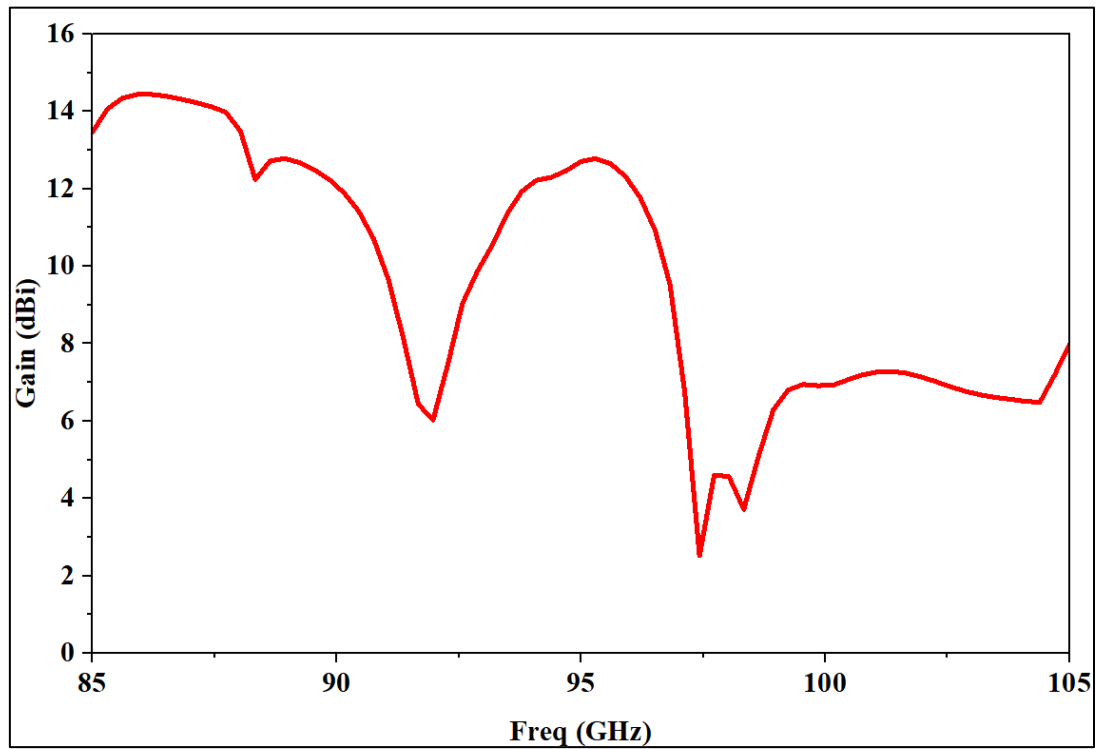


Fig. 15. Gain v frequency curve of 2x2 cross-slot array antenna

The 3-dimensional gain versus frequency plot in figure 16, shows a **gain of 12.45 dBi** of the 2x2 cross-slot array antenna at the resonant frequency of 96.1 GHz.

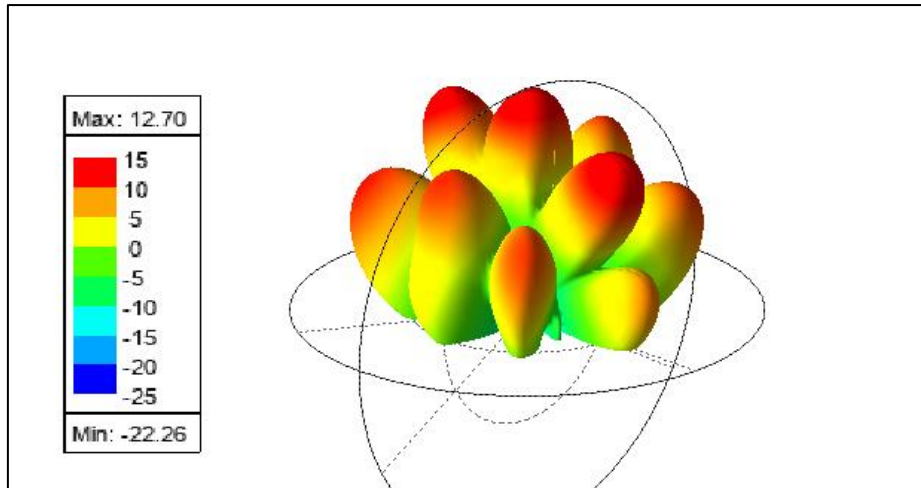


Fig. 16. 3D radiation pattern of 2x2 cross-slot array antenna at 96.1 GHz.

The 3-dimensional radiation pattern in figure 16, shows a **maximum gain of 12.7 dBi** of the 2x2 cross-slot array antenna at the resonant frequency of 96.1 GHz.